Nyx Haile

Quantitative Developer

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Education

Massachusetts Institute of Technology May 2026

B.S. Mathematics, Minor in Economics *Cambridge, MA*

- Coursework: algorithms, theory of computation, computer systems security, cryptography, statistics.

Experience

Partial 2025 – Present

Cofounder & CEO *London*

- Co-founded Partial, a prediction market that turns any relational analysis between assets into a tradable position.
- Built Partial's Rust pricing engines, which assemble any contract from Boolean combinations of a few simpler contracts.
- Designed eleven new classes of tradable contract, from price paths and correlations to lead-lag and reported statistics, all under one pricing framework.

Weiss Laboratory & The SHED, MIT June 2025 – Sept 2025

Undergraduate Researcher (UROP), generative AI for vascularised liver tissue *Cambridge, MA*

- Wrote a parametric vascular-network generator specification for the Cellvivo bioreactor, covering stochastic branching, tapering, R-tree collision avoidance and a 0.99 diffusion-coverage loop.

Selected Projects

Repositories marked private are available on request.

MIT Pokerbots | Python *Heads-up poker-variant bot, January 2026*

- Built a Python bot for MIT Pokerbots' heads-up poker variant; the team placed about 15th of 70.
- Implemented Monte Carlo equity estimation, with exact enumeration for discards, using quick and deep sampling paths to fit a per-decision compute budget.

howtowin.lol | Python, PyTorch, CUDA, Redis *Sequence model for post-game analysis*

- Designed a Dreamer-lineage RSSM sequence model for League of Legends combining static context, player priors and event sequences with outcome, next-event, next-frame and next-decision heads.
- Trained the model end to end on 12.8k games; at minute 15 it predicted the winner at 0.88 AUC on unseen matches and 0.84 on unseen players.

Strand *(private)* | Rust, FoundationDB *Capability-authenticated issue tracker*

- Built a federable, capability-authenticated issue tracker in Rust on FoundationDB, with line-level blame and content-addressed rationale.
- Wrote stable-Rust fuzz harnesses for three frozen v1 formats (wire frames, CBOR citations, backup files); the first pass ran 284,846 iterations with zero crashes.

Adaptive Sparse Core *(private)* | Python, NumPy *Self-rewiring sparse neural model*

- Built a reusable sparse, self-rewiring neural model in Python with a create/train/predict API for classification and regression at any input and output size.
- Implemented temperature-controlled sampled activation, copula-driven rewiring, a loss-plus-sparsity objective and automatic pruning with edge rebalancing.

Forge *(private)* | Rust, systemd, RPC *Local-first agent control plane*

- Built a Rust daemon, operator CLI, Claude Code hooks and provider proxy that coordinate coding-agent work across multiple git repositories.
- Added bounded, jittered exponential backoff to scheduler dispatch, seeded per task so retry waves de-correlate in production and replay exactly in tests.

Technical Skills

Languages: Rust, Python, TypeScript, SQL **ML:** PyTorch, NumPy, CUDA, GPU training and retrieval **Systems:** Linux/systemd, Docker, FoundationDB, PostgreSQL, SQLite, Redis